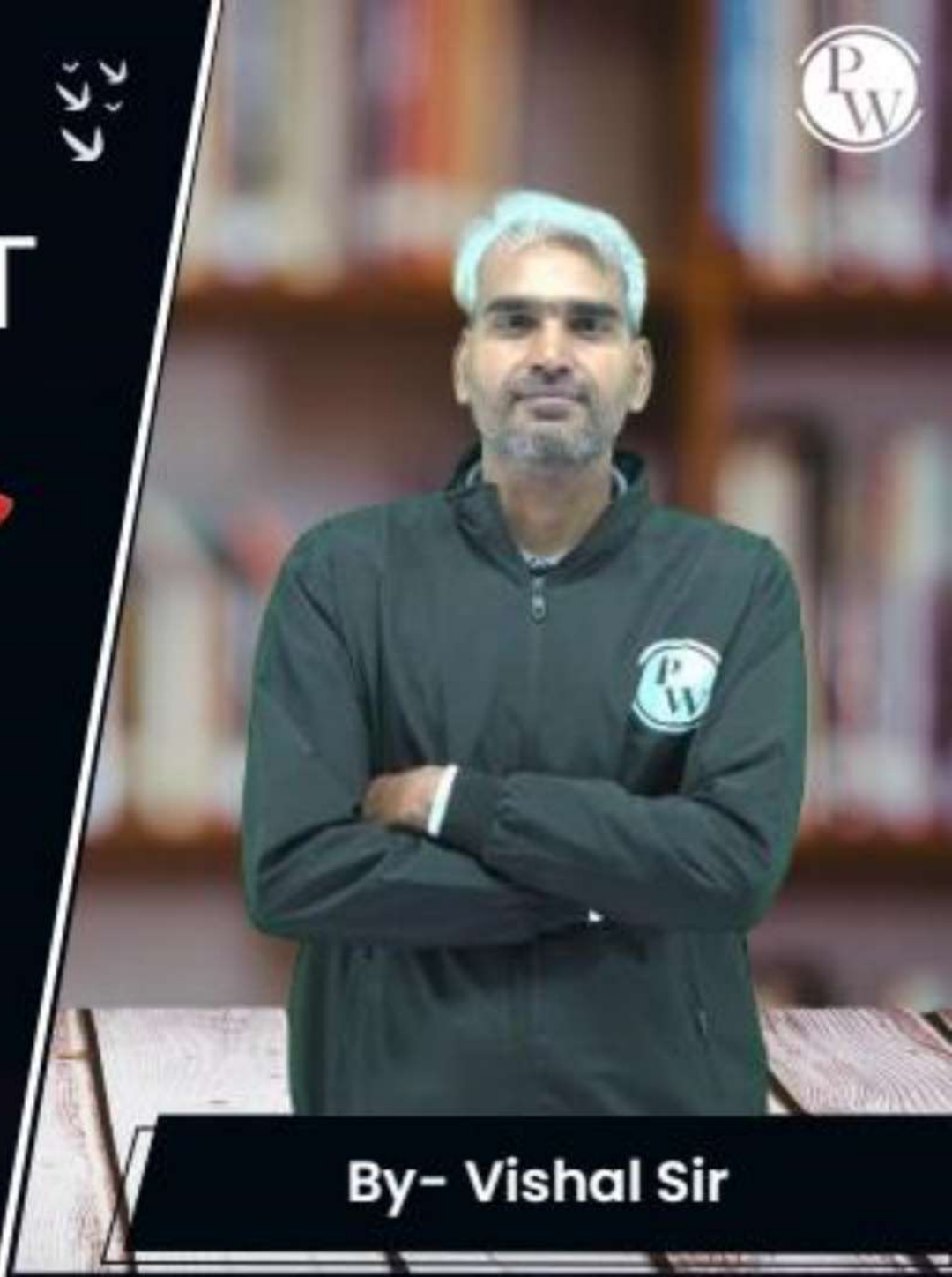
Computer Science & IT



Database Management System

Relational Model & Normal Forms

Lecture No. 10



By- Vishal Sir

Recap of Previous Lecture



Topic Properties of decomposition

Topic Dependency preserving decomposition

Slide

Topics to be Covered

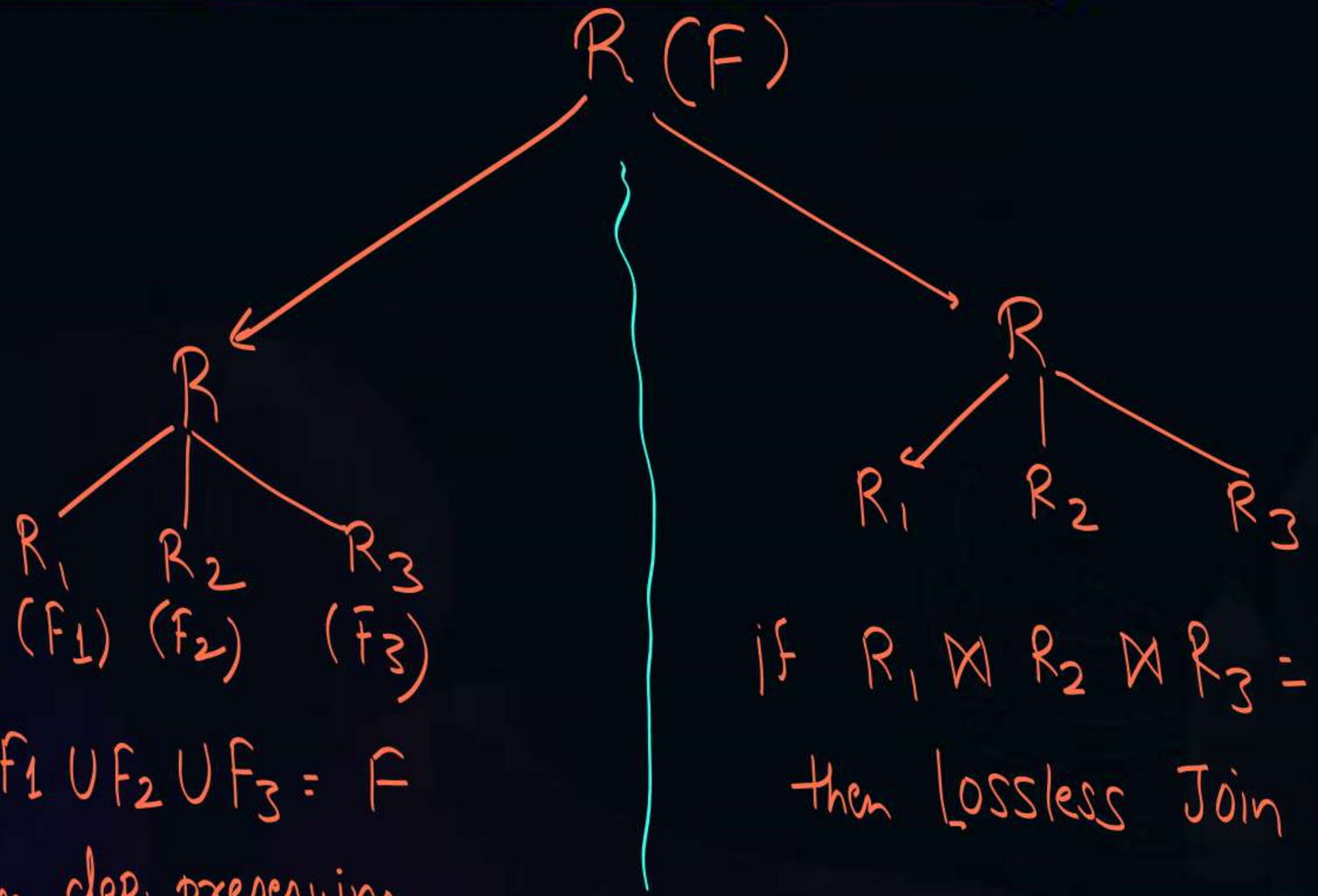


- Topic: Lossless Join decomposition
- Topic: Projection, Selection, and Cross product
- Topic: Natural join
- Topic: Lossless Natural Join

Slide



Topic : Properties of decomposition



if $F_1 \cup F_2 \cup F_3 = F$

then dep. preserving decomposition

if $R_1 \bowtie R_2 \bowtie R_3 = R$

then Lossless Join decomposition



Topic : Lossless Join decomposition

If we decompose a relation R with FD set F into sub-relations R1, R2,.....,Rn with FD sets F1, F2,.....,Fn respectively, then for this decomposition to be called lossless join decomposition following property must hold true.

$$R1 \bowtie R2 \bowtie .... \bowtie Rn = R$$

Natural Join



Topic : Natural Join (⋈)

Natural Join(⋈) is a derived Relational Algebra operation, which is derived using three basic Relational Algebra operation

- Projection (π)
- Selection(σ)
- Cross Product ($\times$)



Topic : Projection (π)

data wrt. specified attributes

It is used to project the column data from a relation based on the attributes specified with projection operation.

Eg: $\pi_{A_1, A_2, \ldots A_n}$ (Name of Relation)

$A_1, A_2, \ldots A_n$: List of attributes from Relation



Topic : Projection (π)

eg:

Enroll

Sid	Cid	Branch
S_1	C_1	CS
S_2	C_2	IT
S_3	C_1	CS

$\pi_{Sid, Cid, Branch}(Enroll) \equiv (Enroll)$

Output will be the Complete "Enroll" table.

$\pi_{Sid}(Enroll) \equiv$

Sid
S_1
S_2
S_3

$\pi_{Cid}(Enroll) \equiv$

Cid
C_1
C_2
C_1 ← duplicate

Note:- Relational Algebra operation will always produce distinct tuples

$\pi_{Sid, Cid}(Enroll) \equiv$

Sid	Cid
S_1	C_1
S_2	C_2
S_3	C_1



Topic : Selection(σ)

It is used to select the tuples(records) from underlying relation based on the predicate condition specified with selection operation.

Eg: $\sigma_{(\text{Predicate Condn})}(\text{Name of Reln})$



Topic : Selection(σ)

Enroll

Sid	Cid	Branch
S_1	C_1	CS
S_2	C_2	IT
S_3	C_1	CS

Retrieve the tuples corresponding to Course = C_1

$\sigma_{(Cid == C_1)}(Enroll) =$

Sid	Cid	Branch
S_1	C_1	CS
S_3	C_1	CS

eg: Retrieve Sids of the students who enrolled for course "C_1"

$= \Pi_{Sid}(\sigma_{(Cid == C_1)}(Enroll))$

Sid
S_1
S_3

$$\Pi_{Sid, Cid, Branch}\left(\sigma_{((Cid == C_1) \vee (Branch = IT))}(Enroll)\right) \Rightarrow$$

Sid	Cid	Branch
S_1	C_1	CS
S_2	C_2	IT
S_3	C_1	CS



Topic : Cross Product (×)

Cross-product is a binary operation. Let R and S are any two relation, then cross product R × S will result in all attributes of R followed by all attribute of S with all possible combinations of tuples from R and S.



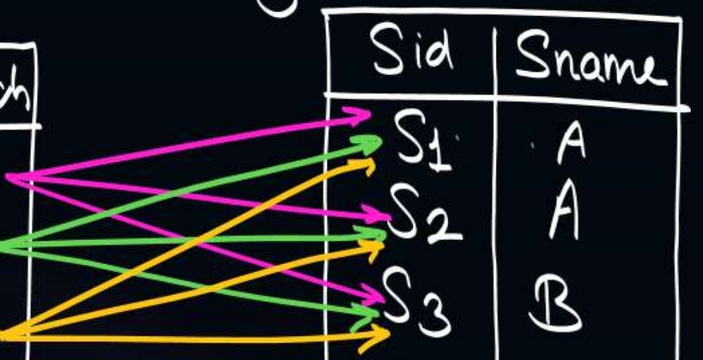
Topic : Cross Product (×)

R =

Sid	Cid	Branch
S_1	C_1	CS
S_2	C_2	IT
S_3	C_1	CS

S =

Sid	Sname
S_1	A
S_2	A
S_3	B



R×S =

R.Sid	R.Cid	R.Branch	S.Sid	S.Sname
S_1	C_1	CS	S_1	A
S_1	C_1	CS	S_2	A
S_1	C_1	CS	S_3	B
S_2	C_2	IT	S_1	A
S_2	C_2	IT	S_2	A
S_2	C_2	IT	S_3	B
S_3	C_1	CS	S_1	A
S_3	C_1	CS	S_2	A
S_3	C_1	CS	S_3	B

- If R has 'x' attributes and S has 'y' attributes, then 'R×S' will contain 'x+y' number of attributes
- If R has 'm' tuples and S has 'n' tuples then 'R×S' will have 'm.n' tuples



Topic : Natural Join (⋈)

Natural join(⋈) is a derived relational algebra operation which is derived using cross product, selection and projection as follows:

Let R and S are any two relations then,

R⋈S = Step-1: Obtain "R × S"

Step-2: Select the tuples from "R × S" based on the equality condition on all common attributes of R and S.

Step-3: Project distinct attributes from the result of step-2.

Topic : Natural Join (⋈)

R =

Sid	Cid	Branch
S_1	C_1	CS
S_2	C_2	IT
S_3	C_1	CS

S =

Sid	Sname
S_1	A
S_2	A
S_3	B

$R \times S =$

R.Sid	R.Cid	R.Branch	S.Sid	S.Sname
S_1	C_1	CS	S_1	A
S_1	C_1	CS	S_2	A
S_1	C_1	CS	S_3	B
S_2	C_2	IT	S_1	A
S_2	C_2	IT	S_2	A
S_2	C_2	IT	S_3	B
S_3	C_1	CS	S_1	A
S_3	C_1	CS	S_2	A
S_3	C_1	CS	S_3	B

Attributes of R

Attributes of S

4 $R \bowtie S = \pi_{R.Sid, R.Cid, R.Branch, S.Sname}\left(\sigma_{(R.Sid = S.Sid)}(R \times S)\right)$

Step-1

Step-2

Sid	Cid	Branch	Sname
S_1	C_1	CS	A
S_2	C_2	IT	A
S_3	C_1	CS	B

① R(Sid, Cid, Branch) , & S(Sid, Sname)

$$R \bowtie S = \pi_{R.Sid,\ R.Cid,\ R.Branch,\ S.Sname}\left(\sigma_{R.Sid = S.Sid}(R \times S)\right)$$

② $R(\underline{A}, \underline{B}, \underline{C})$ & $S(\underline{B}, \underline{C}, \underline{D})$

{ B & C both are Common in R & S }

$$R \bowtie S = \Pi_{R.A, R.B, R.C, S.D}\left(\sigma_{(R.B = S.B) \wedge (R.C = S.C)}(R \times S)\right)$$

③ $R(A, B, C)$ & $S(D, E)$

$R \bowtie S = \pi_{R.A, R.B, R.C, S.D, S.E}(\underbrace{\quad}_{\text{Equality Condn on Common attributes of R \& S}}(R \times S))$

$= R \times S$

If there is no Common attribute in the relations, then natural join will degenerate into Cross Product i.e. $R \bowtie S = R \times S$

↓

Because there is no Common attribute b/w R & S, ∴ there will be no Selection operation



Topic : Lossless Join decomposition

Let relation R is decomposed into sub-relations R1, R2,.....,Rn

In general, $R1 \bowtie R2 \bowtie .... \bowtie Rn \supseteq R$

if, $R1 \bowtie R2 \bowtie .... \bowtie Rn = R$ then, Lossless join decomposition

if, $R1 \bowtie R2 \bowtie .... \bowtie Rn \supset R$ then, Lossy join decomposition

$R1 \bowtie R2 \bowtie .... \bowtie Rn \subset R$ (not possible)

* Consider the following Relation

R =

A	B	C
1	1	1
2	1	2
3	2	1

Let R is decomposed into two sub-relations $R_1(AB)$ & $R_2(AC)$

Check whether the decomposition is lossy or lossless.

Common attribute in R_1 & R_2 is A.

Values of A are unique for all tuples in R_1 as well as in R_2 ∴ lossless

R_1 =

A	B
1	1
2	1
3	2

R_2 =

A	C
1	1
2	2
3	1

$R_1 \times R_2$ =

R_1.A	R_1.B	R_2.A	R_2.C
1	1	1	1
1	1	2	2
1	1	3	1
2	1	1	1
2	1	2	2
2	1	3	1
3	2	1	1
3	2	2	2
3	2	3	1

$R_1 \bowtie R_2$ =

A	B	C
1	1	1
2	1	2
3	2	1

⇒ $R_1 \bowtie R_2 = R$

∴ lossless Join decomposition

* Consider the following Relation.

R =

A	B	C
1	1	1
2	1	2
3	2	1

Let R is decomposed into two sub-relations $R_1(AB)$ & $R_2(BC)$

Check whether the decomposition is lossy or lossless.

Common attribute in R_1 & R_2 is 'B'

values of B are neither Unique in R_1 nor Unique in R_2

∴ lossy

R_1

A	B
1	1
2	1
3	2

R_2

B	C
1	1
1	2
2	1

$R_1 \bowtie R_2$ =

A	B	C
1	1	1
1	1	2
2	1	1
2	1	2
3	2	1

tuples from R

Extra tuples (Spurious tuples)

We can observe

$R_1 \bowtie R_2 \supset R$

∴ lossy Join decomposition

* Consider the following Relation

R:

A	B	C
1	1	2
2	1	2
3	2	1

Let R is decomposed into two sub-relations $R_1(AB)$ & $R_2(BC)$

Check whether the decomposition is lossy or lossless.

Common attribute in R_1 & R_2 is 'B'

values of B are not unique in R_1, but values of B are unique in R_2

↓

lossless

R_1

A	B
1	1
2	1
3	2

R_2

B	C
1	2
2	1

tuple (1, 2) will appear only once

$R_1 \bowtie R_2$ =

A	B	C
1	1	1
2	1	2
3	2	1

$R_1 \bowtie R_2 = R$ ∴ lossless Join decomposition

* Consider the following Relation.

R =

A	B	C
1	1	2
2	1	2
3	2	1

Let R is decomposed into two sub-relations $R_1(AB)$ & $R_2(AC)$

Check whether the decomposition is lossy or lossless.

Common attribute is A, and values of A are unique in R as well as in R_2

∴ lossless

R_1

A	B
1	1
2	1
3	2

R_2

A	C
1	2
2	2
3	1

$R_1 \bowtie R_2$ =

A	B	C
1	1	2
2	1	2
3	2	1

$R_1 \bowtie R_2 = R$

∴ lossless Join decomposition



Topic : NOTE

If relation R is decomposed into two sub-relations R_1 & R_2, then this decomposition will be lossless Join decomposition if and only if following conditions are satisfied.

1. Attributes of R_1 $\cup$ Attributes of R_2 = Attributes of R
2. (Attributes of R_1 $\cap$ Attributes of R_2) $\neq \emptyset$
   - if $\cap = \emptyset$ then Natural Join will degenerate into Cross Product, and it will always be lossy
3. Set of Common attributes b/w R_1 and R_2 must be a Super key of at least one or both the relation

#Q. Let R (A, B, C, D, E) be the relational schema with following FD set

F = { AB→C, C→D, B→E)

Which of the following decomposition is/are lossless join decomposition?

(i){R1(ABC), R2(CD)}

ABC ∪ CD ≠ ABCDE ∴ lossy

(ii){R1(ABC), R2(DE)}

(i) ABC ∪ DE = ABCDE ✓

(ii) ABC ∩ DE = ∅ ∴ lossy

(iii){R1(ABC), R2(CDE)}

(i) ABC ∪ CDE = ABCDE ✓

(ii) ABC ∩ CDE = C ≠ ∅ ✓

(iii) Common attribute = C ⇒ $(C)^+$ wrt F = {C, D}

C is not a Superkey of any one of the Rels.

∴ lossy

(ii){R1(ABCD), R2(BE)}

(i) ABCD ∪ BE = ABCDE ✓

(ii) ABCD ∩ BE = B = ∅ ✓

∴ B is a superkey of R_2

Hence lossless Join

$(B)^+$ wrt F = {B, E}

Both attributes of R_2

Note: ① Natural Join is commutative

i.e. $R_1 \bowtie R_2 = R_2 \bowtie R_1$

② Natual Join is associative

$(R_1 \bowtie R_2) \bowtie R_3 = R_1 \bowtie (R_2 \bowtie R_3)$

H.W.

#Q. Let R (A, B, C, D, E, F) be the relational schema with following FD set

F = { AB→C, BC→A, AC→B, AD→E, B→D, E→F)

Which of the following decomposition is lossless join decomposition?

{ ABC ∪ ABDE ∪ EF } = (All attributes of R)

H.W (i) {R1(ABC), R2(ABDE), R3(EF)}

✓ (ii) {R1(ABC), R2(ADF), R3(ACDE)}

H.W (iii) {R1(AB), R2(BC), R3(ABDE), R4(EF)}

if there exists any order in which relations can be joined s.t. Join is lossless at every Point of Join, then overall decomposition is lossless

(ii) $R_1(ABC)$ $R_2(ADF)$ $R_3(ACDE)$

$R_1 \cap R_2 = A$

$(A)^+ = \{A\}$

Not a s.k of either R_1 or R_2

∴ R_1 & R_2 can not be joined

H.W.

#Q. Let R (A, B, C, D, E, F) be the relational schema with following FD set

F = { AB→C, BC→A, AC→B, AD→E, B→D, E→F)

Which of the following decomposition is lossless join decomposition?

{ ABC ∪ ABDE ∪ EF } = (All attributes of R)

(i){R1(ABC), R2(ABDE), R3(EF)}

✓ (ii){R1(ABC), R2(ADF), R3(ACDE)}

(iii){R1(AB), R2(BC), R3(ABDE), R4(EF)}

if there exists any order in which relations can be joined s.t. Join is lossless at every Point of Join, then overall decomposition is lossless

(ii) $R_1(ABC)$ $R_2(ADF)$ $R_3(ACDE)$

$R_1 \cap R_2 = AC$

$(AC)^+$ = All Attributes

∴ AC is a S.K

∴ We can join R_1 & R_3

$R_1 \bowtie R_3 = (ABCDE)$

$(R_1 \bowtie R_3) \cap R_2 = AD$

$(AD)^+ = \{A, D, E, F\}$

$\therefore$ AD is a S.K of R_2

hence $(R_1 \bowtie R_3)$ can be joined with R_2

$(R_1 \bowtie R_3) \bowtie R_2 = \{A, B, C, D, E, F\}$

lossless Join

lossless Join

Join is lossless at every point of join in the observed order.

$\therefore$ Overall decomposition is lossless.



2 mins Summary

- **Topic** Lossless Join decomposition
- **Topic** Projection, Selection, and Cross product
- **Topic** Natural join
- **Topic** Lossless Natural Join

THANK - YOU